Explainability in Reinforcement Learning

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Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning (RL) still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the Latent Space (LS) learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the Latent Representations (LRs) produced by the Neural Network (NN) and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user's expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representation (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the concept structure of the LS. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

This chapter is organized as follows:

- Section 1.1 lays out the theoretical assumptions underlying the proposed CAR method, which we term the representation convergence hypothesis.
- Section 1.2 then details the CAR method itself, covering both the extraction of LRs using surrogate policies and the automatic detection of concept structures through clustering.
- Section 1.3 finally introduces the metrics we propose for evaluation.

1.1 Representation Convergence Hypothesis

The CAR method relies on a assumption, referred to as the Representation Convergence Hypothesis. This hypothesis follows the two assumptions established previously throughout this manuscript.

The first assumption rejects the notion that deep NNs constitute irreducible black boxes **??**. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be highly complex, they remain explainable through the analysis of the internal mechanisms.

The second assumption states that NNs achieve generalization by learning concepts **??**. During optimization, the network progressively organizes its LS into representations that capture the information required to solve the task.

Following these hypotheses, it becomes possible to explain the NN's process by understanding the organization of its LS. This family of approaches is commonly referred to as mechanistic interpretability.

In mechanistic interpretability, universality refers to the consistency with which a concept is recovered across different models or independent extraction processes [Templeton et al., 2024].

The Representation Convergence Hypothesis provides a theoretical explanation for this phenomenon. We hypothesize that, for a given task and a given NN architecture, there exists a privileged conceptual organization. This organization naturally emerges because it constitutes an efficient representation required to solve the task. Under this assumption, the universality of a concept is not merely an empirical observation but the expected consequence of a common underlying conceptual structure shared by independently learned representations.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a abstract level of representation. We assume that the conceptual organization induced by their LR converges toward an equivalent conceptual structure, even though the numerical implementation of the networks differ.

The remainder of this chapter builds upon this hypothesis. This principle forms the theoretical foundation of the *Clustering Universality* metric, introduced in Subsection 1.3.3.

1.2 Method

As developed in Chapter **??**, several hypotheses have been proposed regarding the organization of LRs in deep NNs. Following the work of [Zahavy et al., 2017], we analyze the LS through clustering techniques, placing our approach within the broader field of Deep Clustering. This design choice is motivated by our rejection of the superposition hypothesis and, more generally, of any view that assumes a linearly organized concept structure, as discussed in Section **??**.

The CAR methodology is composed of two main stages. The first stage trains multiple independent NNs, called surrogate policies, which implement the same decision policy. The second stage clusters their LRs in order to identify the underlying conceptual organization.

This section is organized as follows. In the first subsection, we justify the use of multiple surrogate policies in order to study the stability of representations. In the second subsection, we detail how these surrogate policies are trained so as to best match the original policy.

1.2.1 Multiple surrogate policies

The CAR method was developed with the aim of verifying the representation convergence hypothesis presented in Section 1.1. To test this hypothesis, several NNs must be trained independently and their LRs compared. If similar conceptual structures consistently emerge, this provides empirical evidence supporting the Representation Convergence Hypothesis.

In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. RL introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. Making it impossible to determine whether differences observed in the LS originate from the learning dynamics or from genuine differences between the learned policies.

To overcome this limitation, we consider an already trained initial policy. This policy is then imitated using supervised behavioral cloning. To this end, we independently train several surrogate NNs. Since all surrogate models approximate the same target function, they are converge toward equivalent decision policies. This makes it possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method, as defined in Subsection **??**. The surrogate policies remain deep NNs. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence. The explanations are therefore still extracted directly from the internal representations of deep NNs, making CAR a fully intrinsic and mechanistic explainability approach.

1.2.2 Training surrogate policies

Throughout the remainder of this manuscript, the initial policy we aim to explain is denoted as π. This initial policy is obtained without imposing any constraints on its origin. It may result from classical or deep reinforcement learning [**?**], alpha-beta algorithm [**?**], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π.

The replicated policies are referred to as surrogate policies, denoted by π'. For any observation O, the distribution over the action space produced by π' should approximate that of π as closely as possible.

The surrogate policy π' is implemented as a NN parameterized by Θ. This network is composed of two principal components: an encoder and a decoder, respectively highlighted by the blue and black dashed rectangles in Figure 1.1. The encoder, represented by the function f_{enc} and delimited by the blue rectangle, maps observations from $\mathcal{O}$ into a LS $\mathcal{Z}$. The decoder, delimited by the black rectangle, then reconstructs actions from this LR. Both rectangles encapsulate the embedding layer in the figure, illustrating that the output of the encoder is the input of the decoder.

Figure 1.2 summarizes this overall procedure: the initial policy π is shown

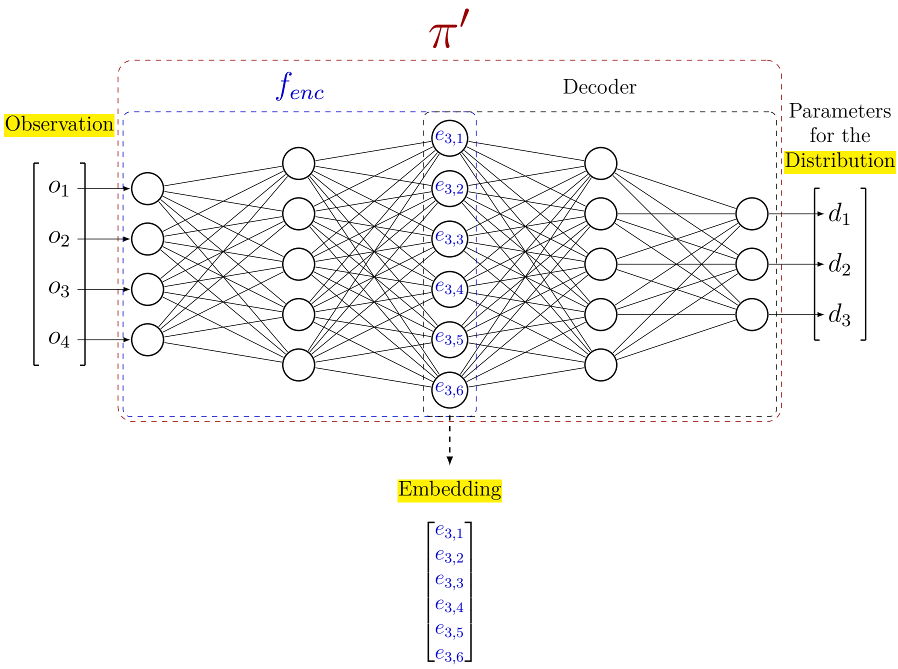


Figure 1.1: Architecture of the surrogate policy π'. The blue dashed box denotes the encoder f_{enc}, which maps the observation O to the embedding, and the black dashed box denotes the decoder, which reconstructs the action distribution parameters from it. Both boxes encapsulate the embedding layer, illustrating that the output of the encoder is the input of the decoder.

as the green box, and the surrogate policy π' as the red box, each producing a distribution over the action space $\mathcal{A}$ from the same observation O. The projection performed by f_{enc} into the LS is shown as the blue crosses.

To train π' to accurately replicate the action distribution, we define a global loss function $\mathcal{L}$ optimized via gradient descent as introduced in Section **??**. This loss, defined in Equation (1.1), is computed as the Mean Squared Error (MSE) between the parameters of the action distributions, obtained by applying the function p, which maps a distribution to its parameterization. In practice, this function does not need to be explicitly defined, as neural policies directly output the parameters of the action distribution. Once training is complete, this loss ensures that π' closely approximates the behavior of the original policy π.

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}\left(O\right) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}\Big(p\big(\pi\left(O\right)\big), p\big(\pi'_{\Theta}\left(O\right)\big)\Big) \tag{1.1}$$

We denote by $\mathcal{E}$ the set of embeddings obtained by applying f_{enc} to a subset of observations $\mathbb{O} \subseteq \mathcal{O}$.

On the set $\mathcal{E}$, we use a clustering function f_{clust} to partition the point cloud of embedded observations into distinct clusters. Each observation is assigned a cluster label based on its proximity in the LS, resulting in the clustering $\mathcal{C}$. Algorithm 1 summarizes the complete CAR procedure, from the behavioral

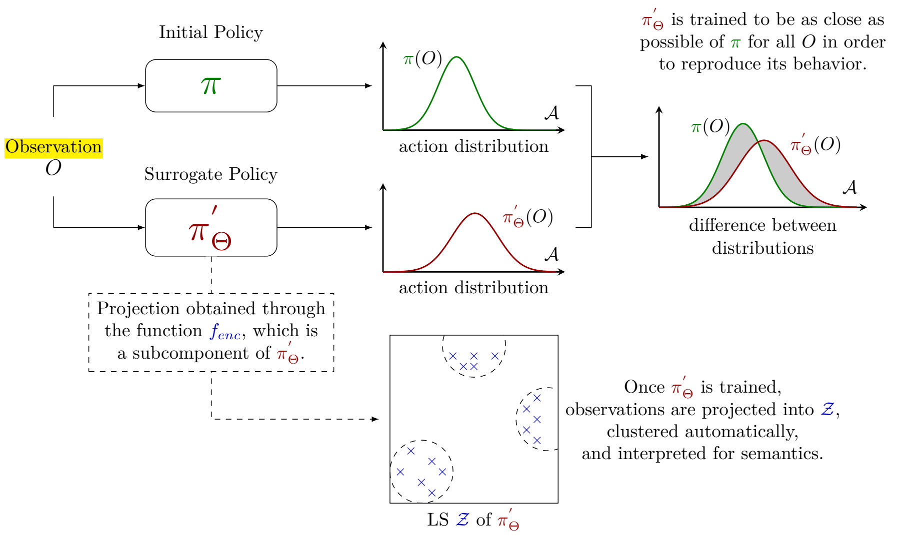


Figure 1.2: Illustration of the Clustering Agent Representation method.

cloning of the surrogate policies to the clustering of their LRs.

To demonstrate this clustering faithfully reflects the agent's internal decision-making process, we introduce a set of evaluation metrics, which will be detailed in the following section.

1.3 Evaluation Metrics

A key contribution of this work is the development of three evaluation metrics, designed to assess the effectiveness of CAR:

- ***Behavior Reconstruction*** : it measures how well a surrogate policy faithfully reproduces the behavior of the original policy by comparing their cumulative rewards.
- ***Clustering Universality*** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- ***Abstraction Score*** : it quantifies the extent to which LS clusters capture abstractions rather than directly reflecting the structure of observations or actions.

The remainder of this section details the motivation for each metric and describes how they are calculated. In the first subsection, we develop the *Behavior Reconstruction*. We then introduce the Adjusted Rand Index (ARI) measure, which serves as the foundation for the two other metrics introduced in the subsequent subsections, namely *Clustering Universality* and *Abstraction Score*.

The following notations will be adopted for the rest of the manuscript: we consider n surrogate policies, denoted as $\pi'_1, \ldots, \pi'_n$, trained as described in

Algorithm 1: Clustering Agent Representation (CAR)

Input:
Original policy π
Observation dataset $\mathcal{O}$
Number of surrogate policies n
Surrogate policy architecture π' (encoder f_{enc}, decoder)
Learning rate $\eta > 0$
Subset of observations $\mathbb{O} \subseteq \mathcal{O}$
Clustering function f_{clust}
Output:
Trained surrogate policies $\pi'_{\Theta_1}, \dots, \pi'_{\Theta_n}$
Clusterings $\mathcal{C}_1, \dots, \mathcal{C}_n$

Behavioral cloning dataset built from the target policy π
$\mathcal{D} \leftarrow \big\{ \big(O, p(\pi(O))\big) \mid O \in \mathcal{O} \big\}$;
MSE loss between action-distribution parameters, as defined in Equation (1.1)
$\mathcal{L} \leftarrow$ MSE;
Stage 1: independently train n surrogate policies by behavioral cloning
for $i = 1, \dots, n$ **do**
  Initialize Θ_i randomly;
  Behavioral cloning of π via SGD, see Algorithm ??
  $\Theta_i \leftarrow \texttt{SGD}(\mathcal{D}, \pi', \mathcal{L}, \eta, \Theta_i)$;
Stage 2: cluster the latent representations of each surrogate policy
for $i = 1, \dots, n$ **do**
  Embeddings produced by the encoder of π'_{Θ_i}
  $\mathcal{E}_i \leftarrow \big\{ f_{enc_i}(O) \mid O \in \mathbb{O} \big\}$;
  Partition of the embedded observations into clusters
  $\mathcal{C}_i \leftarrow f_{\text{clust}}(\mathcal{E}_i)$;
Return $\big(\pi'_{\Theta_1}, \dots, \pi'_{\Theta_n}\big), \big(\mathcal{C}_1, \dots, \mathcal{C}_n\big)$

Section 1.2. Each surrogate policy is associated with a projection function $f_{enc_1}, \dots, f_{enc_n}$, which is applied to the set of observations $\mathbb{O}$, yielding embeddings $\mathcal{E}_1, \dots, \mathcal{E}_n$. A clustering function is then applied to each embedding set, producing partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$.

1.3.1 Behavior Reconstruction

We argue that explaining a function b that closely approximates another function a is essentially the same as explaining a. If b accurately captures the key decision patterns of a, understanding b provides meaningful insights into how a works [Guidotti et al., 2018]. In our setting, this means that generating explanations for π' that faithfully reproduce π amounts to explaining π directly. To verify that π' is indeed a faithful copy, we introduce the *Behavior Reconstruction* metric.

This metric is computed by comparing the cumulative rewards obtained by the original policy π and the surrogate policies π' over m evaluation episodes.

The behavior reconstruction metric is then defined as:

$$\mathit{Behavior\ Reconstruction} = \frac{1}{n\sum_{i=1}^{m} g_{\pi}^{i}} \sum_{j=1}^{n}\sum_{k=1}^{m} g_{\pi'_j}^{k} \tag{1.2}$$

In imitation learning, it is unlikely to expect the surrogate policy π' to outperform the initial policy π it aims to approximate through supervised learning [Ross and Bagnell, 2010]. Consequently this metric ranges in $[0, 1]$. Values approaching 1 indicate minimal loss in reproducing π, while values close to 0 indicate poor reproduction.

We deliberately adopt a reward-based definition rather than alternatives based on actions or action distributions (e.g., training loss, accuracy, KL divergence [Shlens, 2014], or optimal transport [Peyré and Cuturi, 2020]). Since surrogate policies are trained in a supervised manner, they cannot develop strategies beyond those demonstrated by the initial policy. This means that if the rewards achieved are comparable, the surrogate policy is effectively copying the original policy's behavior.

1.3.2 Adjusted Rand Index

Within CAR, concepts are assumed to correspond to clusters in the LS. Combined with the Representation Convergence Hypothesis, this leads to a direct experimental prediction: if independently trained surrogate policies converge toward the same conceptual organization, then clustering their LRs should produce similar partitions.

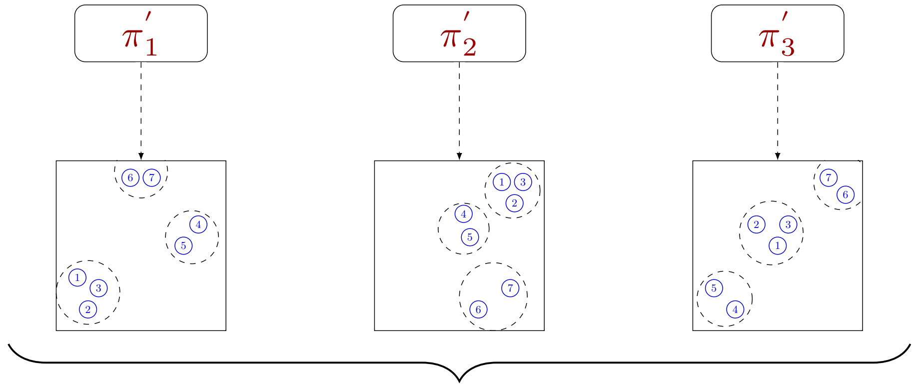
When a particular observation is processed by the network, it is expected to activate the same concept regardless of the specific neural implementation of the policy. If independently trained policies converge toward the same conceptual organization, then a given observation should activate the same concept across all models. As a consequence, observations grouped together within a cluster for one policy should also be grouped together for another independently trained policy.

In the clustering literature, the most used measures for comparing two partitions is the ARI [Hubert and Arabie, 1985]. This measure quantifies the agreement between cluster assignments independently of the spatial organization of the clusters. Both metrics introduced later in this chapter, namely *Clustering Universality* and *Abstraction Score*, rely on the ARI. Since the ARI is a complex measure, we review its definition and discuss its main properties before introducing our proposed metrics.

The ARI is derived from the Rand Index (RI) [Rand, 1971], which measures the agreement between two partitions by considering every possible pair of point cloud. For each pair, four situations are possible: the point may belong to the same cluster in both partitions, to different clusters in both partitions, or they may disagree by being grouped together in only one of the two partitions. The RI measures the proportion of pairs for which the two partitions agree:

$$\mathrm{RI} = \frac{\text{number of agreeing pairs}}{\text{total number of pairs}}.$$

Although the RI is intuitive, it suffers from an important limitation. In most clustering problems, the majority of pairs of points belong to different clusters.



$$Clustering\ Universality = \frac{1}{\binom{n}{2}} \sum_{i=1}^{n} \sum_{j=0}^{i-1} \mathrm{ARI}(\mathcal{C}_i, \mathcal{C}_j)$$

Figure 1.3: Three point clouds obtained by projecting the same observations through three independently trained surrogate policies π'_1, π'_2, and π'_3. The numbered circles 1 to 7 denote the same seven observations across the three point clouds, and the dashed contours denote the clusters recovered in each latent space.

Consequently, even two completely independent partitions will agree on a large number of pairs, simply because both assign them to different clusters. This effect becomes more pronounced as the number of points or clusters increases.

The ARI addresses this issue by accounting for the agreement that would be expected purely by chance. It compares the observed agreement RI to the expected agreement under random partitions $\mathbb{E}[\mathrm{RI}]$. The score is then normalized by the maximum agreement achievable beyond this random baseline, $\max(\mathrm{RI}) - \mathbb{E}[\mathrm{RI}]$. As a result, an ARI of 0 corresponds to the level of agreement expected from random clusterings, while an ARI of 1 indicates perfect agreement between the two partitions:

$$\mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{\max(\mathrm{RI}) - \mathbb{E}[\mathrm{RI}]}$$

In practice, the ARI is computed directly from the cluster assignments of the two partitions. Let clustering A be composed of partitions $\{A_i\}$ and clustering B of partitions $\{B_j\}$. Let $a_i = |A_i|$ denote the number of samples in partition i of clustering A, and $b_j = |B_j|$ the number of samples in partition j of clustering B. Furthermore, let $n_{ij} = |A_i \cap B_j|$ be the number of samples shared by partition i of clustering A and partition j of clustering B, and let n be the total number of samples. With these definitions, the ARI can be computed as:

$$\mathrm{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}$$

Figure 1.3 illustrates this property of the ARI. It shows three point clouds obtained by projecting the same seven observations, labeled 1 to 7, through three independently trained surrogate policies. Although the numbered points occupy different positions within each point cloud, the dashed clusters group exactly the same elements in the three cases: the partitioning is therefore identical. In particular, point 1 represents the same object in the three point clouds, but appears at a different location because it is projected differently by each surrogate policy. This spatial discrepancy has no effect on the ARI, since the measure only depends on which cluster each point is assigned to, regardless of where the points are placed in space.

1.3.3 Clustering Universality

Having introduced the ARI, we can now define the *Clustering Universality* metric. While the ARI quantifies the similarity between two conceptual partitions, the objective of CAR is to assess the consistency of conceptual organizations across an entire set of independently trained surrogate policies.

To this end, the same set of observations is projected into the LS of each surrogate policy and clustered independently. Under the Representation Convergence Hypothesis, these independently obtained partitions are expected to be similar. *Clustering Universality* quantifies this by computing the mean pairwise ARI:

$$\textit{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^{n} \sum_{j=0}^{i-1} \mathrm{ARI}(\mathcal{C}i, \mathcal{C}j). \tag{1.3}$$

This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate policies produce highly similar partitions, suggesting that the clusters reflect intrinsic properties of the policy. Conversely, a value near 0 implies that the similarity between partitions is no better than that of two randomly generated clusterings, meaning that the clustering fails to capture any meaningful structure in the policy. A negative value, on the other hand, signals that the agreement between partitions is worse than what would be expected by chance, indicating a systematic disagreement between the clusterings.

1.3.4 Abstraction Score

If *Clustering Universality* is high, it becomes important to determine what underlies this stability. Two possibilities arise:

(i) the LS merely reproduces the structure of the observation space or action distribution, offering no abstraction, as defined in Definition **??**;

(ii) the LS captures similarities in decision-making processes, yielding behaviorally meaningful clusters.

To distinguish among these cases, we introduce the *Abstraction Score*. This metric contextualizes latent clusterings by examining how their patterns align with reference structures derived from the raw observation and action data. We consider : $\mathcal{C}_{obs}$, the clustering in the observation space $\mathbb{O}$, and $\mathcal{C}_{act}$, the clustering in the action-space distribution parameters. The clusterings $\mathcal{C}_{obs}$ and $\mathcal{C}_{act}$ are performed using the same algorithm, distance metric, and hyperparameters as those used for $\mathcal{C}_i$ in the LS, ensuring that any biases introduced by the clustering procedure itself are minimized.

For each latent clustering $\mathcal{C}_i$, we define its similarity to these baselines using the ARI:

$$A_i^{obs} = \text{ARI}(\mathcal{C}_i, \mathcal{C}_{obs})$$
$$A_i^{act} = \text{ARI}(\mathcal{C}_i, \mathcal{C}_{act})$$

The *Abstraction Score* is subsequently calculated as one minus the average of the maximum absolute similarity values across these two reference baselines:

$$\textit{Abstraction Score} = 1 - \tfrac{1}{n} \textstyle\sum_{i=1}^{n} \max\left(\left|A_i^{obs}\right|, \left|A_i^{act}\right|\right) \tag{1.4}$$

The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the latent clusters are strongly aligned with either the observation or the action space, reflecting minimal abstraction, cases (i). A score close to 1 indicates that the latent clusters are independent of both the observation and action spaces, capturing abstractions that meaningfully reflect the agent's behavior, case (ii).

The following chapter presents the results of applying these metrics to specific examples.

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